

Observation Theory for Estimation and Control: Consumer-Induced Geometry as a Complement to Observability, Filtering, and Feedback

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Abstract

Classical control theory separates several questions that are easy to conflate: whether a plant can be driven, whether its state can be inferred, how uncertainty should be filtered, and how actions should be chosen. A further question is usually supplied externally by the designer: *which distinctions in the estimated state matter to the downstream consumer?* Observation Theory (OT) supplies a candidate mathematical object for that missing interface. If a consumer $\mathcal{C} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is judged in a local output metric G , its pullback $P_{\mathcal{C}}(x) = J_{\mathcal{C}}(x)^\top G(\mathcal{C}(x)) J_{\mathcal{C}}(x)$ induces the local state geometry the consumer can distinguish, and estimator uncertainty Σ and consumer geometry compose as the operational uncertainty $U_{\mathcal{C}} = \text{tr}(P_{\mathcal{C}}\Sigma)$. This paper defines a research campaign around that composition, organized as three interfaces to the classical stack: *estimator geometry* (a state-dependent $P_{\mathcal{C}}(x)$ moves the optimal estimate off the posterior mean), *resource value* ($V_{\mathcal{C}} = \text{tr}(P_{\mathcal{C}}(\Sigma^- - \Sigma^+))$, the consumer-relative value of a measurement), and *dynamic relevance* ($S_{\mathcal{C}}(\Delta) = \text{tr}[P_{\mathcal{C}}\Sigma(t+\Delta | I_t)]$, the age of operationally relevant uncertainty). We first establish a boundary: for a fixed positive-semidefinite $P_{\mathcal{C}}$ and an unconstrained Bayesian estimator, the posterior mean — hence the Kalman estimate — remains optimal; both this invariance and the rank-one value-of-observation identity are machine-checked in Lean 4/Mathlib. Each pillar of the program has close prior art in isolation: black-box recovery of quadratic task metrics (decision-focused learning), and consumer-derived weights composed with covariances (LQG sensing co-design, goal-oriented experimental design, value-of-information scheduling) in analytic form. The narrow novelty hypothesis is their conjunction: that a downstream consumer’s operational metric can be recovered from the consumer itself under *query access only*, composed with classical estimator uncertainty, and validated prospectively, out of sample, at matched budgets that include the cost of probing. OT is proposed not as a replacement for observability, filtering, stability, or feedback, but as a complementary theory of operational distinguishability at their downstream boundary.

Index Terms

Kalman filtering, sensor scheduling, value of information, task-oriented communication, goal-oriented experimental design, functional observability, operational distinguishability.

I. THE MISSING QUESTION IN THE CLASSICAL CONTROL STACK

A modern estimation-and-control stack can be schematized as

$$\text{plant} \rightarrow \text{sensors} \rightarrow \text{estimator} \rightarrow \text{consumer(s)} \rightarrow \text{actions}.$$

Classical systems theory provides precise concepts for controllability and observability. Kalman filtering [1] provides an optimal state estimate and error covariance under its linear-Gaussian assumptions. LQR, LQG, MPC, and stochastic control determine actions once an objective has been specified. Yet a design choice remains embedded in these methods: *which state errors actually matter?* In LQR this appears as the choice of a state penalty Q ; in estimation it appears in accuracy specifications, sensor priorities, covariance weights, update rates, and communication policies. In many systems these choices are engineered successfully — but their operational geometry is externally specified.

Observation Theory begins one step downstream. Let $\mathcal{C} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be a consumer and let D_Y be a twice-differentiable divergence on its output. Define the output metric and consumer Jacobian

$$G(z) = \nabla_v^2 D_Y(z, v)|_{v=z}, \quad J_{\mathcal{C}}(x) = \frac{\partial \mathcal{C}}{\partial x}(x), \quad (1)$$

and the *read operator*

$$P_{\mathcal{C}}(x) = J_{\mathcal{C}}(x)^\top G(\mathcal{C}(x)) J_{\mathcal{C}}(x). \quad (2)$$

For sufficiently small state error $\delta = x - \hat{x}$,

$$D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x})) \approx \frac{1}{2} \delta^\top P_{\mathcal{C}}(x) \delta, \quad \mathbb{E} D_Y \approx \frac{1}{2} \text{tr}(P_{\mathcal{C}} \Sigma_{\delta}), \quad (3)$$

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This manuscript extends the author’s Observation Theory program from compression and coding to estimation and control. Program repository, sealed preregistrations, claims ledger, and the Lean 4/Mathlib formalization of Proposition 1 and Proposition 2 are at <https://github.com/ahb-sjsu/geometric-observation>.

so ordinary reconstruction error is the isotropic corner $P_C = I$. The composition

$$U_C = \text{tr}(P_C \Sigma) \quad (4)$$

reads: *what the estimator does not know*, weighted by *what the consumer can distinguish*. Neither object substitutes for the other: an accurate consumer model cannot estimate the plant; a perfect error covariance cannot say which residual uncertainty matters operationally.

The systems question is therefore not “can Observation Theory replace the Kalman filter?” It is: *given a classical estimator that tells us what we do not know, can a consumer-induced geometry tell us which parts of that uncertainty matter — and therefore where scarce sensing, communication, computation, and control resources should be spent?*

A. Three interfaces, not one

The program is organized around three distinct interfaces between the read operator and the classical stack:

$$P_C(x) \longrightarrow \begin{cases} \textbf{Estimator geometry} & \text{if } P_C \text{ varies with state (§III, Rem. 2),} \\ \textbf{Resource value} & \text{tr}(P_C \Delta \Sigma) \quad (\text{§IV–§V}), \\ \textbf{Dynamic relevance} & \text{tr}(P_C \Sigma(\Delta)) \quad (\text{§IX, conjectural}). \end{cases} \quad (5)$$

Interface 1 is *opened* rather than closed by the boundary theorem of §III: the posterior-mean invariance holds for fixed P_C , and its failure under state-dependent $P_C(x)$ is precisely where consumer geometry can change the estimator itself. Interface 2 is the paper’s spine. Interface 3 is developed as an explicitly conjectural branch.

B. Access models: demonstrations, gradients, queries

The distinction that defines this program’s niche is the access model. Inverse optimal control and inverse reinforcement learning recover objectives from *demonstrations*. Decision-focused learning differentiates through a *known* (or differentiable) consumer [24]–[26]. Observation Theory probes a *black-box* consumer under a declared observation distribution. Only the third access model is available when the consumer is a learned perception stack, a physical link, or another party’s system — and those consumers, which resist hand-modeling, are exactly where a recovered geometry is necessary rather than convenient.

II. OBSERVABILITY IS NOT OPERATIONAL DISTINGUISHABILITY

Consider $x_{k+1} = Ax_k + Bu_k + w_k$, $y_k = Cx_k + v_k$. Classical observability asks whether distinct internal states can be distinguished from the measurement history. OT asks a downstream question: for consumer $\mathcal{C}$, two nearby states x and $x + \delta$ are *locally operationally distinguishable* according to $\delta^\top P_C(x) \delta$. A direction in $\ker P_C$ is locally invisible to that consumer; a direction with a large eigenvalue of P_C is strongly read. Hence a state direction can be perfectly observable to the sensor system and nearly irrelevant to the consumer; conversely, a very small estimation error can be operationally dominant if it lies in a strongly weighted read direction. These are complementary geometries on opposite sides of the estimator.

Two classical structures must be engaged here by name. First, *functional observability* [2], [3] asks when a hand-specified linear functional $z = Lx$ can be reconstructed; OT’s linear special case $\mathcal{C}(x) = Lx$ lies inside its subject matter, and OT adds the *metric* (not merely subspace) structure, nonlinear consumers, and recovery of L -like structure from the consumer itself. Second, for a linear consumer $z = Lx$ with output metric G , the pointwise OT geometry is $P_C = L^\top G L$, and the output-weighted observability Gramian of frequency-weighted balanced truncation [4] is its *dynamics-propagated accumulated analogue*,

$$W_C(T) = \int_0^T e^{A^\top t} P_C e^{At} dt. \quad (6)$$

Classical observability geometry thus *emerges from accumulated OT geometry* in the linear case: the Gramian is what the consumer’s instantaneous read metric integrates to under the flow.

III. A BOUNDARY PROPOSITION: OT DOES NOT CREATE A NEW KALMAN FILTER

The first result of the campaign is deliberately negative.

Proposition 1 (Weighted posterior-mean invariance). *Let X be a random state, Y the available observations, and $P \succeq 0$ a fixed state weighting. Then $\mathbb{E}[(X - \hat{x})^\top P (X - \hat{x}) \mid Y]$ is minimized over $\hat{x}$ by the posterior mean $\hat{x} = \mathbb{E}[X \mid Y]$.*

Proof. Let $m = \mathbb{E}[X \mid Y]$. Then $X - \hat{x} = (X - m) + (m - \hat{x})$, the cross terms vanish in conditional expectation, and

$$\mathbb{E}[(X - \hat{x})^\top P (X - \hat{x}) \mid Y] = \text{tr}(P \Sigma_{X|Y}) + (m - \hat{x})^\top P (m - \hat{x}),$$

whose second term is nonnegative. □

The posterior mean already minimizes *every* fixed positive-semidefinite quadratic state loss; in the linear-Gaussian case this is the Kalman estimate. If the campaign merely rediscovered a weighted Kalman filter, it would not constitute the proposed contribution. The opportunity lies in deciding which measurements to take, which information to transmit, which state directions to retain, when to update, and eventually which operational errors to regulate — and, per Remark 2, in the state-dependent regime the invariance itself fails.

Remark 1 (Machine-checked form; weakened hypotheses). Proposition 1 (discrete form) and Proposition 2 below are machine-checked in Lean 4 against Mathlib (`lean/ObservationTheory/WeightedMeanInvariance.lean`; Lean and Mathlib v4.32.2, clean build, no `sorry`). The formalization shows the invariance survives under hypotheses weaker than stated: only $v^\top P v \geq 0$ for all v and weights summing to one are used — neither symmetry of P nor nonnegativity of the weights. We state the natural $P \succeq 0$ form above; the weakening is recorded here because the kernel checked it, not because the generality is operationally needed. Per the program’s charter, mechanical verification is necessary but not sufficient for a `[proved]` ledger classification, which additionally requires an independent fresh-context derivation pass.

Remark 2 (State dependence opens Interface 1). Proposition 1 is for *fixed* P . For a nonlinear consumer, $P_C(x)$ depends on x ; the loss $\mathbb{E}[(X - \hat{x})^\top P_C(X)(X - \hat{x}) \mid Y]$ is no longer a fixed quadratic, and its minimizer is in general *not* the posterior mean. Consumer geometry can then change the estimator itself, not only the allocation of resources around it. Characterizing this estimator shift is an open problem of the campaign, not a result of this paper.

IV. CONSUMER-RELATIVE VALUE OF OBSERVATION

Let a Bayesian or Kalman estimator hold prior covariance Σ^- and consider a candidate measurement $y = Hx + v$, $v \sim \mathcal{N}(0, R)$. The covariance update gives $\Sigma^+ = \Sigma^- - \Sigma^- H^\top (H \Sigma^- H^\top + R)^{-1} H \Sigma^-$. Define the *consumer-relative value of observation*

$$V_C = \text{tr}(P_C(\Sigma^- - \Sigma^+)) = \text{tr}(P_C \Sigma^- H^\top (H \Sigma^- H^\top + R)^{-1} H \Sigma^-). \quad (7)$$

Proposition 2 (Rank-one form). *For a scalar measurement $y = h^\top x + v$ with $\text{Var}(v) = r$ and a rank-one consumer $P_C = gg^\top$,*

$$V_C = \frac{(g^\top \Sigma^- h)^2}{h^\top \Sigma^- h + r}. \quad (8)$$

This form is the section’s message: a sensor is not operationally valuable because it observes a high-variance direction; it is valuable when its measurement direction h , *propagated through the current uncertainty* Σ^- , overlaps a direction g the consumer actually reads. The covariance algebra is classical. The proposed OT contribution is the *provenance* of g or P_C : supplied by the downstream consumer’s distinguishability geometry rather than chosen as an engineering weight.

V. RESOURCE-CONSTRAINED OPERATION IS THE NATURAL INTERFACE

Weighted covariance objectives are established in sensor selection and scheduling [9], [12] and in LQG sensing co-design [6]; the optimization $\min_a \text{tr}(M \Sigma^+(a))$ for a designer-specified M is not an OT novelty. The proposed pipeline is

$$\text{consumer} \xrightarrow{\text{probe}} \widehat{P}_C \xrightarrow{\text{compose}} \text{tr}(\widehat{P}_C \Sigma) \xrightarrow{\text{allocate}} a^*,$$

with the allocation judged on the actual downstream consumer.

A. Probe cost is part of the objective

Recovering $\widehat{P}_C$ — and re-recovering it along trajectories when the consumer is nonlinear — consumes exactly the resources OT allocates. The policy problem is therefore not $\min_\pi D_C(\pi)$ subject to side constraints, but

$$\min_\pi D_C(\pi) + \lambda_s C_{\text{sensing}}(\pi) + \lambda_c C_{\text{comm}}(\pi) + \lambda_p C_{\text{probing}}(\pi), \quad (9)$$

which contains a genuine exploration problem: *when is another consumer query worth its cost?* The marginal value of a probe — how much a further refinement of $\widehat{P}_C$ changes downstream allocations — is itself an OT quantity. All campaign comparisons in §XI are run under (9), with probe cost on the same budget line as sensing and communication.

B. Greedy selection: a stated limitation

Weighted-trace objectives $\text{tr}(P_C \Delta \Sigma)$ are *not* supermodular in general: the counterexamples of Jawaid and Smith [10] cover trace-of-covariance objectives, so greedy selection under (7) does not inherit the classical $(1 - 1/e)$ guarantee that log-determinant objectives enjoy. We do not attempt to repair this here. Available machinery includes approximate-supermodularity bounds [11] and convex relaxations [9]; establishing *conditions under which V_C -greedy selection retains approximation guarantees* is registered as a campaign theorem — a named target, not a claimed result.

C. Operational event triggering

An operational trigger updates when the uncertainty visible to the consumer becomes excessive — $\text{tr}(P_C \Sigma_t) > \tau$ — rather than when a timer expires or isotropic variance crosses a threshold. The analytic special case is the value-of-information trigger of Soleymani, Baras, and Hirche [13], [14], in which the transmit rule derived from the downstream LQG cost is provably optimal. Any OT triggering claim must cite that line as its analytic ancestor and either match it (LQG) or generalize beyond it (non-analytic consumers).

VI. CONSUMER GEOMETRY AS A LOCAL CONTROL GEOMETRY

If the ultimate objective is expressed at a downstream output $z = \mathcal{C}(x)$ with desired value $z^* = \mathcal{C}(x^*)$, a second-order expansion gives $D_Y(\mathcal{C}(x), \mathcal{C}(x^*)) \approx \frac{1}{2}(x - x^*)^\top P_C(x^*)(x - x^*)$, so $P_C(x^*)$ is a natural local state penalty — a candidate *provenance* for one component of an LQR/MPC objective, not a replacement for it. For a nonlinear consumer the penalty is trajectory dependent. Stability, robustness, terminal costs, actuator constraints, and feasibility remain control-theoretic problems; the proposed division of labor is that OT supplies where the state penalty comes from, and control theory determines what to do with it.

A. Threshold consumers and the belief-averaged read operator

The local metric (2) presumes a Hessian. Threshold-like consumers — bit-error rate, outage, any cliff — have degenerate or explosive G near the threshold, and the paper’s own application exemplar (§XIII) is of this type. Where the local geometry exists, the robustified object is the *belief-averaged read operator*

$$\bar{P}_C(b) = \mathbb{E}_{X \sim b} [J_C(X)^\top G(\mathcal{C}(X)) J_C(X)], \quad (10)$$

where b is the estimator’s belief: the read operator is averaged over the uncertainty the estimator actually has, rather than evaluated at a point the system may not occupy. For genuinely nonsmooth outage metrics no Hessian exists to average, and a finite-perturbation or smoothed operational metric must replace (2) rather than pretending one does. *Belief-relative Observation Theory* — the read operator as a functional of the belief — is flagged as a substantial extension in its own right.

VII. BLACK-BOX RECOVERY IS THE SHARP CLAIM

If P_C must always be hand-specified, much of this program reduces to established task-weighted estimation and control. The stronger proposition is that the read geometry can, under stated conditions, be identified from the consumer itself: for appropriate consumers, finite-difference probes under a declared observation distribution recover sensitivity information without privileged access to the consumer’s implementation. The probing *methodology* is itself prior art — empirical observability Gramians recover plant sensitivity by perturb-and-simulate [5] — and in machine learning, locally optimized decision losses recover quadratic task-metric surrogates from a black-box decision oracle by sampled perturbations [21], [22]. What is claimed here is the probed *object* (a downstream consumer’s operational metric), its *composition* with an explicit estimator covariance, and the *validation protocol*: recovery on one perturbation distribution followed by success only on the same perturbations is insufficient; the test must be out of sample, at matched budgets, on the held-out consumer.

VIII. MULTIPLE CONSUMERS

A real state service rarely has one downstream consumer. Given $P_1, \dots, P_m$ for navigation, tracking, collision avoidance, communications, or mission logic, a weighted sum $P_C^{\text{tot}} = \sum_i \lambda_i P_i$ can be useful but can hide incompatibility. Observation Theory’s two-observer rate-distortion result supplies a precedent: in the Gaussian weighted-quadratic problem, successive refinement is lossless exactly when the observers’ optimal residual uncertainty ellipsoids nest; distinct observers can demand geometrically incompatible information. That result is a coding theorem, not yet a control theorem. The corresponding systems question — when can one estimator, sensor schedule, or state representation efficiently serve several consumers, and when does their operational geometry force a multi-consumer tax? — remains to be discovered, and must not be answered by smuggling the coding result across.

IX. DYNAMIC RELEVANCE: THE AGE OF OPERATIONALLY RELEVANT UNCERTAINTY

Interface 3 is stated as an explicit conjecture. Let $\Sigma(t+\Delta \mid I_t)$ be the uncertainty predicted from currently retained information. Define

$$S_C(\Delta) = \text{tr}[P_C(t+\Delta) \Sigma(t+\Delta \mid I_t)]. \quad (11)$$

An observation becomes stale when the uncertainty it leaves *in the consumer’s read directions* becomes operationally expensive — age of information [31], [32] weighted by operational geometry rather than by the clock. Our prior-art sweep found no anisotropic-staleness formulation; the nearest neighbor is the analytic (scalar-LQG) value-of-information line [13]. Because (11) has its own community, baselines (AoI, AoII, VoI), and falsifier (does directional staleness beat age-based scheduling at matched update budgets on a real consumer?), it is retained here only as a conjectural branch and pursued independently. If long-horizon consequences dominate immediate consumer-relative covariance reduction, S_C may remain a feature or terminal metric rather than a sufficient scheduling policy; that outcome would be a scientific boundary, not a failure to be patched.

X. PRIOR ART AND THE NOVELTY HYPOTHESIS

The foundational claim must be narrower than several established areas, and — following a dedicated three-slice sweep (control theory; machine learning; communications and the application domain) — narrower still than the original draft of this paper stated. Each pillar of the hypothesis exists separately in strong form. Local quadratic task metrics are recovered from black-box decision oracles by sampled probing in decision-focused learning [21]–[23]. Consumer-derived weights composed with estimator covariances drive prospective sensor and bit allocation in LQG sensing co-design [6], goal-oriented Bayesian experimental design [18]–[20] — including Jacobian/Hessian pullbacks of nonlinear goal maps with adjoint access — value-of-information scheduling [13]–[15], rate-cost control [16], [17], goal-oriented quantization [28], and task-based quantization [29]. Attention and anticipation select sensing resources against task-projected covariance metrics in closed loop [7]; task-driven representations are co-designed by information bottlenecks [8]; consumer-derived perceptual metrics already steer codec bit allocation in practice [30]. No novelty is claimed for any of these, nor for Kalman filtering, posterior-mean optimality, PSD-weighted covariance objectives, sensor selection and scheduling, active sensing, semantic communication, loss-calibrated inference [27], or inverse optimal control. The analytic LQG case is conceded entirely.

Research hypothesis (conjunction). A downstream consumer induces an operational state geometry $P_C = J^\top G J$ that can, under explicit observation conditions, be recovered from the consumer itself under *query access only* and a declared observation distribution; composed with classical estimator uncertainty as $\text{tr}(P_C \Sigma)$; and used to predict and optimize the operational value of sensing, communication, representation, and local control resources for consumers *outside the analytic families* — validated prospectively, out of sample, at matched budgets that include the cost of probing, on the actual downstream consumer.

This is a novelty hypothesis, not a novelty verdict: no single work found combines black-box probed consumer geometry, explicit covariance composition, and prospective matched-budget validation, but a referee holding decision-focused learning in one hand and goal-oriented experimental design in the other could reasonably call the conjunction an integration. The answer is the integration *plus the validation protocol* — reconstruction-matched flips, held-out consumers, matched budgets, physical endpoints — which no found work runs.

XI. RESEARCH CAMPAIGN

Campaign 1 — Verify the boundary: Random linear systems; direct optimization over admissible linear estimator gains compared with the Kalman solution under randomly drawn PSD consumer metrics. Any repeatable improvement over the Kalman estimate indicates an implementation error or a model violation. The invariance is classical and now machine-checked (Remark 1); the campaign *verifies* the boundary publicly rather than discovers it.

Campaign 2 — Consumer-relative value of observation: A dynamical state with heterogeneous candidate sensors and a planted low-rank consumer; selection by (7). The signature experiment is a *reconstruction-matched flip*: two sensing policies with essentially identical $\text{tr} \Sigma$ but differently shaped uncertainty; only the consumer changes. Preregistered prediction: $\text{tr}(P_C \Sigma)$ identifies which policy is better for the downstream consumer while ordinary state MSE cannot. Where an analytic model exists, baselines must include the Riccati/VoI-derived weight — not merely hand-tuned or isotropic alternatives.

Campaign 3 — Blind recovery, then scheduling (flagship): The point at which the pieces stop resembling analytic LQG/VoI machinery:

unknown consumer $\rightarrow$ query-only recovery of $\widehat{P}_C \rightarrow$ sensor scheduling $\rightarrow$ held-out real consumer,

at equal sensing budgets with probe cost included per (9), and with an *analytic-consumer positive-control arm*: on a consumer whose optimal weight is known (LQG/VoI), the blind pipeline is required to *recover and match* the known optimum — not beat it. Matching the analytic optimum blind is the calibration proof; winning on non-analytic consumers is the contribution. Success requires more than subspace recovery: the recovered operator must choose useful measurements and improve the held-out consumer at matched budget.

Campaign 4 — Operational event triggering: Limited transmissions or measurement energy; compare periodic, age-based, isotropic-covariance, hand-weighted, analytic-VoI (where available), and recovered-OT triggering. Primary outcome: downstream consumer performance at matched communication or energy — never lower OT surrogate loss by construction.

Campaign 5 — Physical consumer: A moving RF or free-space optical tracking system (§XIII). Navigation state includes position, velocity, attitude, angular rate, and structural motion; the physical consumer reads angle error, optical coupling, received power, BER, or outage. The key test is again reconstruction-matched: two navigation systems with similar state RMSE but different covariance geometry should produce different physical performance, and a blindly recovered $\widehat{P}_C$ should predict the ordering *before* the run.

Campaign 6 — Multi-consumer state service: Two consumers with read subspaces ranging from aligned to orthogonal; compare common estimation, weighted scalarization, separate service, and layered/refined descriptions. The two-observer coding theorem supplies hypotheses about incompatibility but must not be imported as a control result; the goal is to determine whether a genuinely dynamic analogue of observer complementarity exists.

Campaign 7 — Closed-loop control: Only after the estimation and sensing claims survive: a recovered or derived $P_C(x)$ inside an LQR/MPC objective against isotropic and competently hand-tuned baselines at matched control effort, with stability and constraint satisfaction independently audited, and the endpoint again the real downstream consumer.

XII. FALSIFIERS

The program narrows or fails if: the recovered read operator does not generalize outside its probing distribution; a conventional engineer-specified task matrix performs as well, making recovery operationally unnecessary; $P_C(x)$ varies too rapidly for a local metric to hold over realistic estimator uncertainty (the belief-averaged operator (10) is the first line of defense, and its failure would bound belief-relative OT as well); probe cost under (9) consumes the advantage the recovered geometry provides; or long-horizon closed-loop consequences systematically defeat policies based on consumer-relative covariance reduction, in which case OT remains a theory of consumer-relative representation with local and terminal uses. These outcomes are scientific boundaries, to be reported, not patched away.

XIII. APPLICATION EXEMPLAR: COOPERATIVE OPTICAL HIGH-ALTITUDE PLATFORMS

A useful application — not the foundation of the theory — is a constellation of persistent high-altitude platforms connected by free-space optical links [33]. Each vehicle has a navigation and attitude state; each optical terminal reads only particular combinations of transverse displacement, line-of-sight angle, attitude, angular velocity, structural jitter, range, and optical state. Its physical consumer geometry need not resemble 6-DOF navigation MSE, and the standard practice in the pointing-acquisition-tracking literature is a static top-down pointing-error budget: the causal arrow runs forward (attitude error $\rightarrow$ link budget) only. The OT proposal runs it backward: probe the black-box receiver through modeled state perturbations, recover a belief-averaged read operator (10) — the endpoint metrics are threshold-like, so the smoothed construction of §VI-A is required, not optional — and contract it with the platform’s Kalman covariance to decide when to request a beacon update, which navigation measurement to acquire, which information to transmit to a neighbor, and how much precision to allocate to each state direction. Multiple simultaneous links provide a natural multi-consumer test. The advantage of this exemplar is that its endpoint — received power, fiber coupling, BER, outage — is external and physical: OT cannot win merely by optimizing its own metric.

XIV. CONCLUSION

Observation Theory proposes an estimation-and-control research program centered on a compact composition, $U_C = \text{tr}(P_C \Sigma)$: the covariance belongs to estimation theory, the read operator to the consumer’s operational geometry; the former says what is uncertain, the latter what matters. Three interfaces organize the program — estimator geometry when P_C varies with state, resource value through $\text{tr}(P_C \Delta \Sigma)$, and dynamic relevance through $S_C(\Delta)$ — and none replaces observability, filtering, LQG, MPC, or stability theory. The campaign succeeds only if its sharpest claim survives: that P_C can be recovered from real downstream consumers under query access, at accounted probe cost, and prospectively predicts resource allocations that ordinary state reconstruction does not — matching the analytic optimum where one exists and winning where none does. The program is judged by reconstruction-matched flips, blind recovery, held-out dynamics, physical endpoints, competent classical baselines, and explicit negative results. If those gates fail, Observation Theory remains a theory of consumer-relative representation. If they pass, operational distinguishability becomes a working complement to the classical control stack.

REFERENCES

- [1] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, 1960.
- [2] M. Darouach, “Existence and design of functional observers for linear systems,” *IEEE Trans. Autom. Control*, vol. 45, no. 5, pp. 940–943, 2000.
- [3] A. N. Montanari, C. Duan, L. A. Aguirre, and A. E. Motter, “Functional observability and target state estimation in large-scale networks,” *Proc. Natl. Acad. Sci. USA*, vol. 119, no. 1, e2113750119, 2022.
- [4] D. F. Enns, “Model reduction with balanced realizations: An error bound and a frequency weighted generalization,” in *Proc. IEEE Conf. Decision and Control*, 1984, pp. 127–132.
- [5] A. J. Krener and K. Ide, “Measures of unobservability,” in *Proc. IEEE Conf. Decision and Control*, 2009, pp. 6401–6406.
- [6] V. Tzoumas, L. Carlone, G. J. Pappas, and A. Karaman, “LQG control and sensing co-design,” *IEEE Trans. Autom. Control*, vol. 66, no. 4, pp. 1468–1483, 2021.
- [7] L. Carlone and S. Karaman, “Attention and anticipation in fast visual-inertial navigation,” *IEEE Trans. Robotics*, vol. 35, no. 1, pp. 1–20, 2018.
- [8] V. Pacelli and A. Majumdar, “Task-driven estimation and control via information bottlenecks,” in *Proc. IEEE Int. Conf. Robotics and Automation*, 2019, pp. 2061–2067.
- [9] S. Joshi and S. Boyd, “Sensor selection via convex optimization,” *IEEE Trans. Signal Process.*, vol. 57, no. 2, pp. 451–462, 2009.
- [10] S. T. Jawaid and S. L. Smith, “Submodularity and greedy algorithms in sensor scheduling for linear dynamical systems,” *Automatica*, vol. 61, pp. 282–288, 2015.
- [11] L. F. O. Chamon, A. Amice, and A. Ribeiro, “Approximately supermodular scheduling subject to matroid constraints,” *IEEE Trans. Autom. Control*, vol. 67, no. 3, pp. 1384–1396, 2022.
- [12] L. Ye *et al.*, “A unified approach to optimally solving sensor scheduling and sensor selection problems in Kalman filtering,” arXiv:2304.02692, 2023.
- [13] T. Soleymani, J. S. Baras, and S. Hirche, “Value of information in feedback control: Quantification,” *IEEE Trans. Autom. Control*, vol. 67, no. 7, pp. 3730–3737, 2022.
- [14] T. Soleymani, J. S. Baras, S. Hirche, and K. H. Johansson, “Value of information in feedback control: Global optimality,” *IEEE Trans. Autom. Control*, vol. 68, no. 6, pp. 3641–3647, 2023.

- [15] T. Soleymani, J. S. Baras, and K. H. Johansson, "Foundations of value of information: A semantic metric for networked control systems tasks," arXiv:2403.11927, 2024.
- [16] V. Kostina and B. Hassibi, "Rate-cost tradeoffs in control," *IEEE Trans. Autom. Control*, vol. 64, no. 11, pp. 4525–4540, 2019.
- [17] T. Tanaka, K. K. Kim, P. A. Parrilo, and S. K. Mitter, "Semidefinite programming approach to Gaussian sequential rate-distortion trade-offs," *IEEE Trans. Autom. Control*, vol. 62, no. 4, pp. 1896–1910, 2017.
- [18] A. Attia, A. Alexanderian, and A. K. Saibaba, "Goal-oriented optimal design of experiments for large-scale Bayesian linear inverse problems," *Inverse Problems*, vol. 34, no. 9, 095009, 2018.
- [19] K. Wu, P. Chen, and O. Ghattas, "An offline-online decomposition method for efficient linear Bayesian goal-oriented optimal experimental design," *SIAM J. Sci. Comput.*, vol. 45, no. 1, pp. B57–B77, 2023.
- [20] "Goal-oriented optimal experimental design using quadratic approximations of nonlinear goal functionals," arXiv:2411.07532, 2025 (J. Sci. Comput., to appear).
- [21] S. Shah, K. Wang, B. Wilder, A. Perrault, and M. Tambe, "Decision-focused learning without decision-making: Learning locally optimized decision losses," in *Advances in Neural Information Processing Systems*, 2022.
- [22] S. Shah, B. Wilder, A. Perrault, and M. Tambe, "Leaving the nest: Going beyond local loss functions for predict-then-optimize," in *Proc. AAAI Conf. Artificial Intelligence*, 2024.
- [23] D. Bansal, R. T. Q. Chen, M. Mukadam, and B. Amos, "TaskMet: Task-driven metric learning for model learning," in *Advances in Neural Information Processing Systems*, 2023.
- [24] A. N. Elmachet and P. Grigas, "Smart 'predict, then optimize'," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.
- [25] B. Amos and J. Z. Kolter, "OptNet: Differentiable optimization as a layer in neural networks," in *Proc. Int. Conf. Machine Learning*, 2017.
- [26] J. Mandi *et al.*, "Decision-focused learning: Foundations, state of the art, benchmark and future opportunities," *J. Artif. Intell. Res.*, vol. 80, pp. 1623–1701, 2024.
- [27] S. Lacoste-Julien, F. Huszár, and Z. Ghahramani, "Approximate inference for the loss-calibrated Bayesian," in *Proc. Int. Conf. Artificial Intelligence and Statistics*, 2011.
- [28] H. Zou, C. Zhang, S. Lasaulce, L. Saludjian, and H. V. Poor, "Goal-oriented quantization: Analysis, design, and application to resource allocation," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 42–54, 2023.
- [29] N. Shlezinger, Y. C. Eldar, and M. R. D. Rodrigues, "Hardware-limited task-based quantization," *IEEE Trans. Signal Process.*, vol. 67, no. 20, pp. 5223–5238, 2019.
- [30] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition*, 2018.
- [31] Y. Sun, Y. Polyanskiy, and E. Uysal, "Sampling of the Wiener process for remote estimation over a channel with random delay," *IEEE Trans. Inf. Theory*, vol. 66, no. 2, pp. 1118–1135, 2020.
- [32] A. Maatouk, M. Assaad, and A. Ephremides, "The age of incorrect information: An enabler of semantics-empowered communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2621–2635, 2023.
- [33] Y. Kaymak, R. Rojas-Cessa, J. Feng, N. Ansari, M. Zhou, and T. Zhang, "A survey on acquisition, tracking, and pointing mechanisms for mobile free-space optical communications," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1104–1123, 2018.